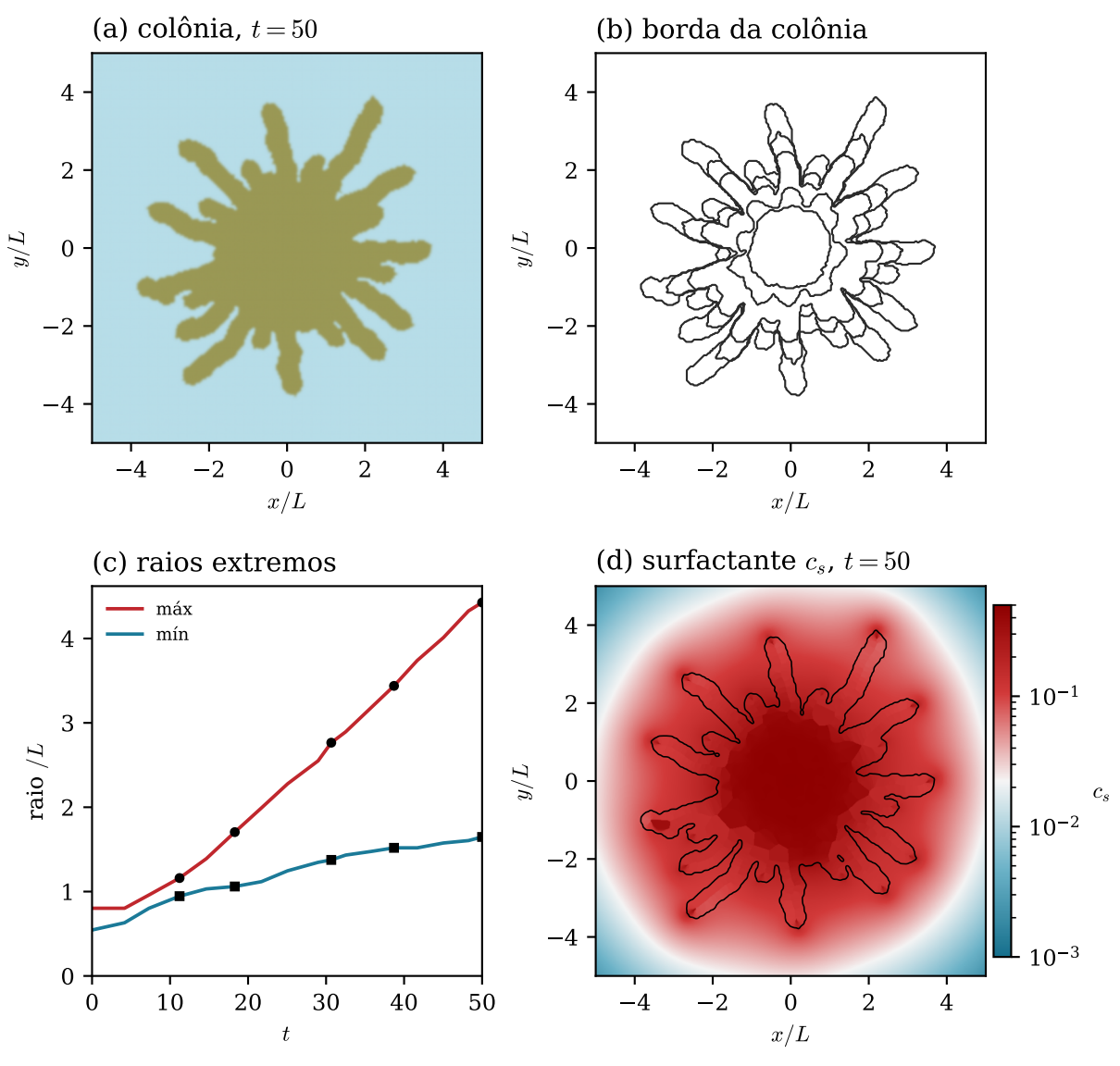

(a) colônia, $t = 50$
(b) borda da colônia
(c) raios extremos
máx
mín
raio /L
t
(d) surfactante c_s, $t = 50$
c_s
10^{-1}
10^{-2}
10^{-3}
x/L
y/L